



EEE3094S 2024: Class Test 2 Memo

- The test is open book. You do not need to hand in your resources.
- **The supporting information for all questions and formula sheet are attached at the end of this paper. You may remove them for easy access.**
- Answer on the test sheet. If you run out of space, you may attach additional pages, but please note that you have done so (e.g. write “see attached pages” by the question) and make sure they are marked with your student number.
- You may do your diagrams in pencil but please ensure that your lines are clear.
- There are 3 questions worth a total of 53 marks. The maximum mark you can score for the test is 50 though.
- You have 90 minutes to complete the test + 15 minutes of reading time.

Question 1 [20 marks]

Godzilla and Mothra are rival students in a kaiju control systems class. They have each come up with a controller design for the same system, and both think that theirs is the best! The Nichols and inverse Nichols charts on the next two pages compare the open-loop frequency responses of the plant with each of their controllers. The uncontrolled plant is second-order and stable in open-loop.

Use the given plots to determine whose design will give better performance in each of the categories below. In some cases, the answer might be that their designs are equivalent. State your choice, and give an explanation supported by features of the plots. You may wish to annotate the plots to make your answers clearer.

- A) Closed-loop tracking of step inputs [3 marks]
- B) Robustness to phase disturbances [3 marks]
- C) Robustness to fluctuations in the parameters of the plant [3 marks]
- D) Gain margin [3 marks]
- E) Reducing closed-loop oscillation [3 marks]
- F) Closed-loop settling time [5 marks]

Solutions:

- A) Godzilla. Larger low frequency gain = better tracking accuracy.
- B) Mothra. The phase margin (x axis distance to the -1 point) is larger, so more phase deflection can occur without causing the system to become unstable.
- C) Equivalent/ Godzilla marginally better. Neither design crosses into the large amplifying circles on the inverse Nichols chart, with the maximum closed-loop value of the sensitivity function being between 1 and 2 dB.
- D) Equivalent. Both systems have infinite gain margin.
- E) Mothra. The larger phase margin indicates a larger damping ratio.
- F) Godzilla's controller is faster because the real part of the closed loop pole is further from the origin.

The real part of the pole is given by $-\zeta\omega_n$, so we need to approximate the damping ratio and natural frequency of the closed-loop system.

Godzilla:

Using $\zeta = \frac{A}{100} \Phi_m$:

The open-loop low-frequency gain for Godzilla's controller is around 27.5 dB, or 23.7. This gives a closed-loop gain of $\frac{23.7}{1+23.7} = 0.95$

The phase margin is around 55 degrees.

The damping ratio is therefore $\zeta \approx \frac{0.95}{100}(55) = 0.53$

The natural frequency happens when the gain is $-6 - 20 \log_{10}(\zeta)$ below the closed-loop DC gain, which here is at around -0.93 dB. The response intersects the -1 dB circle at around 1 rad/s, so we will use this to approximate ω_n . This puts the closed-loop pole at $-(0.53)(1) = -0.53$.

Mothra:

Applying the same method for Mothra:

Open-loop gain = 15 dB = 5.6

$\therefore$ closed loop gain = 0.85.

Phase margin = 70 degrees.

$$\therefore \zeta \approx \frac{0.85}{100}(70) = 0.6$$

$$20 \log_{10}(0.85) - 6 - 20 \log_{10}(0.6) \approx -3 \text{ dB.}$$

The intersection with the -3 M circle happens at around 0.5 rad/s, so $\omega_n = 0.5$.

The real part of the pole is then at $-(0.6)(0.5) = -0.3$.

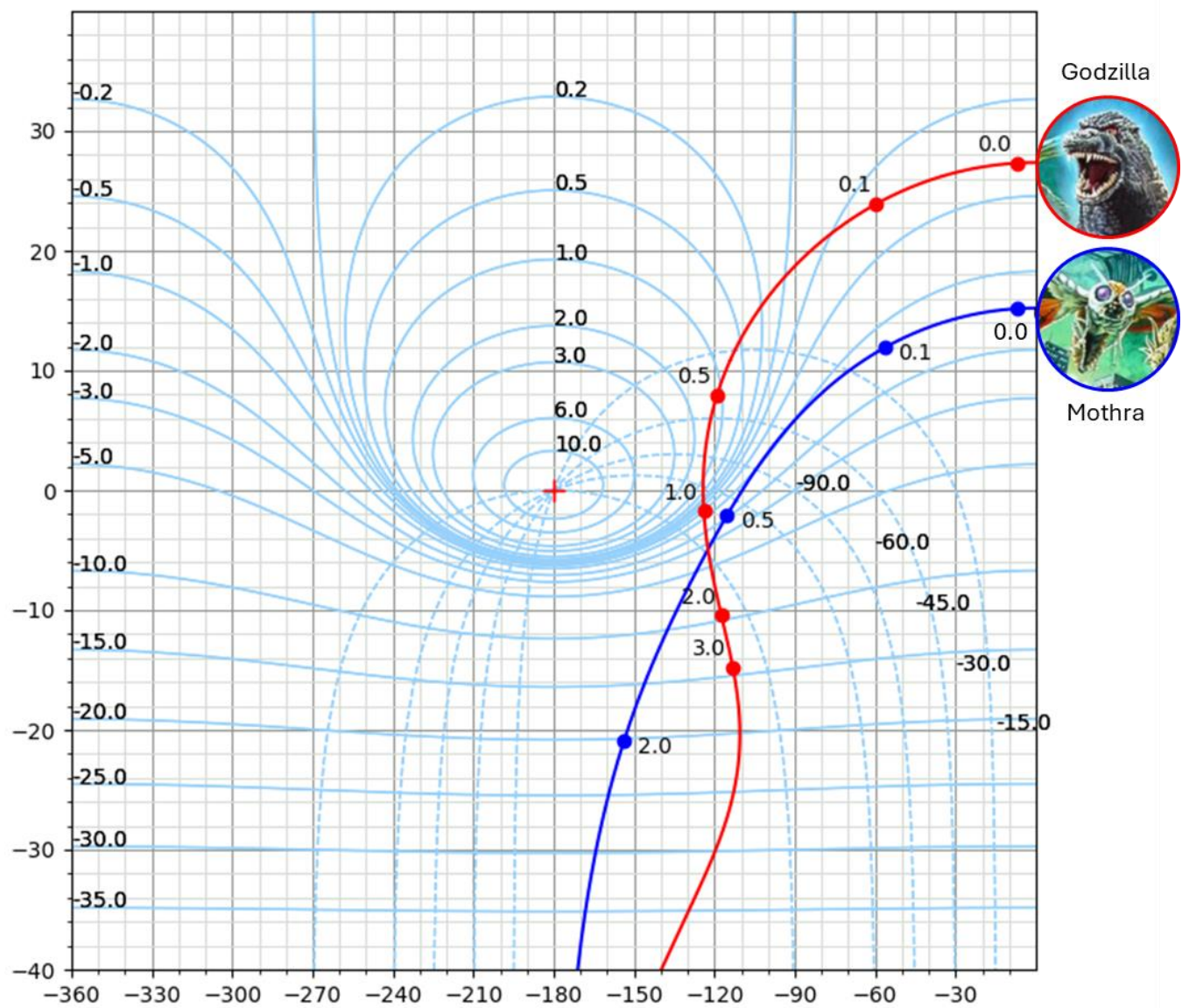


Figure 1: Nichols chart

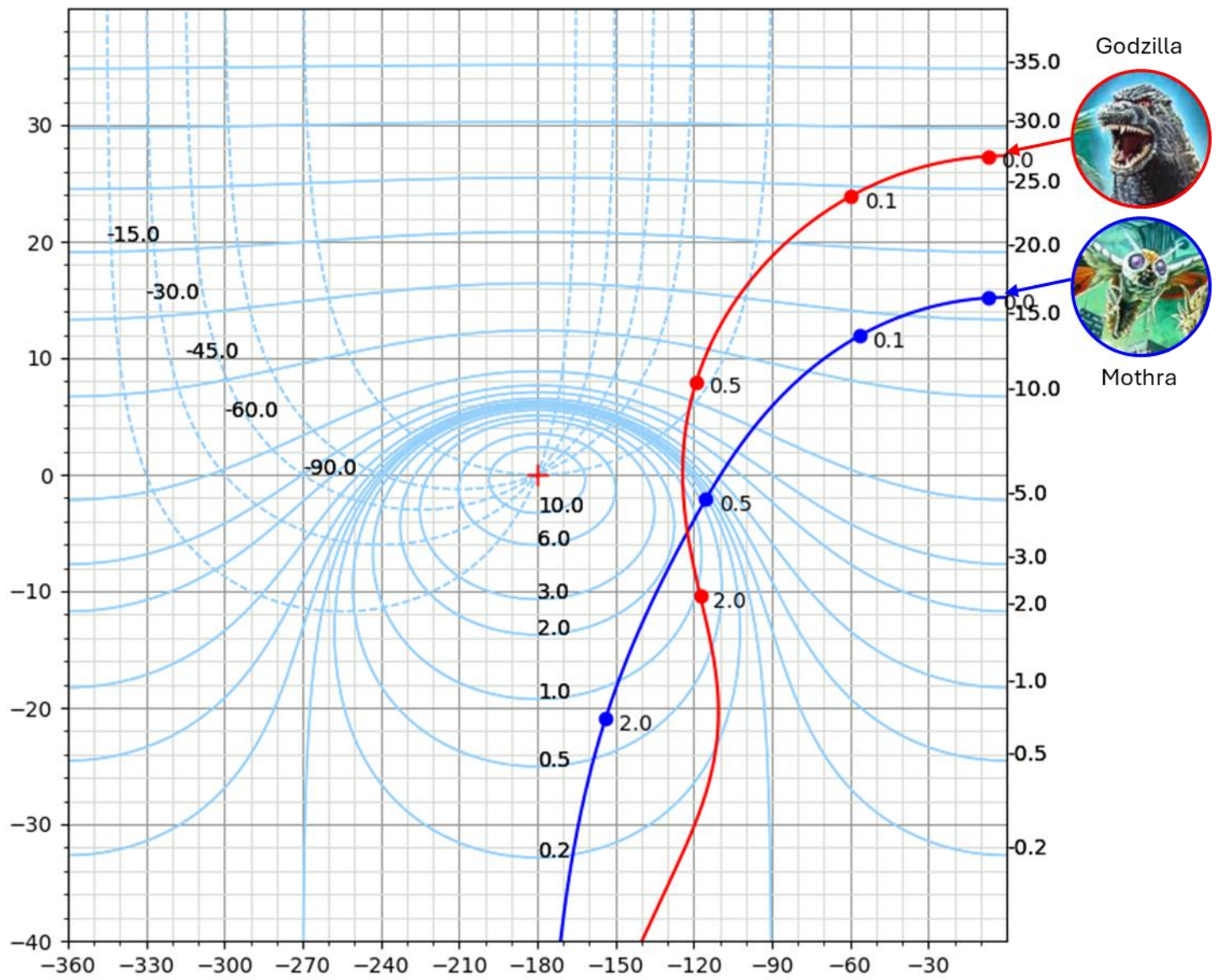


Figure 2: Inverse Nichols chart

Question 2 [15 marks]

Consider the transfer function:

$$G(s) = \frac{0.25}{s(2s + 1)}$$

You might recognize this as the reduced-order transfer function we came up with representing a drone in the last test.

a) You will now be using it to design a proportional controller for the altitude of the drone. When the drone is placed in negative feedback with your controller, the closed-loop system should satisfy the following specifications:

Accuracy:

- Tracking of step inputs to within an error of less than 5%.
- Tracking of ramp inputs with an error of less than 0.1 meters.

Transient response:

- Settling time to within 2% of the final value in under 20 seconds.
- Overshoot of less than 20%.

Your design should include an analysis of these specifications (what is required to fulfil them? Is this possible?) and clearly-described steps (described in words and calculations) showing how you decide on the right gain K for your controller. If it is not possible to meet all the specifications, you should prioritize **transient response**, and calculate the best possible outcomes for the specifications that cannot be met.

Use the root locus below to help you with your design. You may annotate it as needed to explain your working. [12 marks]

b) Is it possible to improve tracking by increasing the type number of the system further? Why/ why not? Hint: you may want to sketch a rough root locus of the system with increased type number to support your answer. [3 marks]

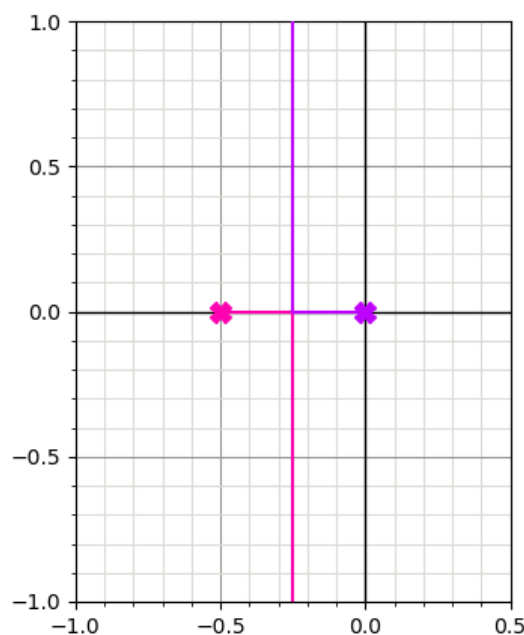


Figure 3: root locus

Solution:

a) Analysis of Specifications:

Accuracy:

The system will track a position setpoint with perfect accuracy, regardless of the controller, as it has one integrator in the transfer function. This can also be concluded using the final value theorem. (2)

Applying the final value theorem of the error for ramp inputs: (2)

$$e(\infty) = \lim_{s \rightarrow 0} s \frac{1}{s^2} \frac{1}{1+KG(s)} = \lim_{s \rightarrow 0} \frac{1}{s+KG(s)} = \lim_{s \rightarrow 0} \frac{1}{s+\frac{K \cdot 0.25}{2s+1}} = \frac{1}{0.25K} \leq 0.1 \therefore K \geq \frac{1}{(0.25)(0.1)} \therefore K \geq 40$$

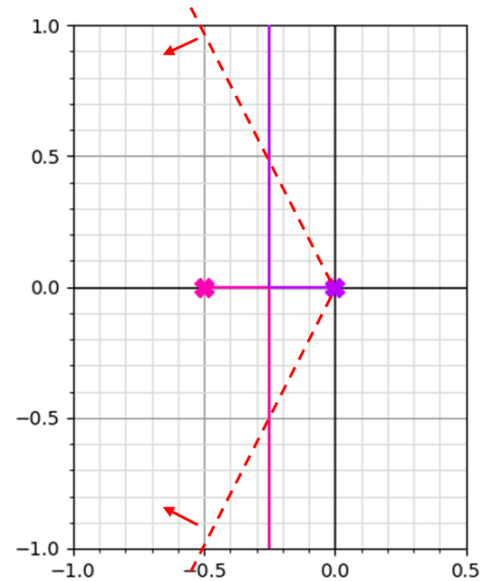
Transient Response:

The settling time to 2% is roughly 4τ , so if $4\tau \leq 20$ s, $\tau \leq 5$ s, so the pole must be further left than $-\frac{1}{5}$. As long as the poles are somewhere on the asymptote at -0.25, this will be satisfied. (2)

Using the formula, $0.2 = e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}}$, we can solve for $\zeta \geq 0.46$. (1)

This specification creates a feasible region of the s plane bounded by lines with angles $\pm \arccos(0.46) = \pm 63^\circ$. (1)

By inspection of the root locus with these boundaries drawn on, we can see that the poles on the boundary are at $-0.25 \pm j0.5$, which is probably close enough to work with.



Alternatively, this can be calculated using the known real part of the pole and the known angle to find $im = 0.25 \tan(63^\circ) = 0.48$, giving $-0.25 \pm j0.48$. (1)

The gain producing this closed-loop pole is found using $K = \frac{1}{|G(s)|}$. Substituting $-0.25 + j0.48$ into $G(s)$ gives -0.43 , so $K = \frac{1}{0.43} = 2.34$. (1)

Gain selection:

Clearly, the gain required to track a ramp to the specified accuracy is much higher than the gain that satisfies the transient response specifications. As instructed, we will prioritize transient response and set the gain to $K = 2.34$. (1)

Applying the final value theorem to the error for a ramp: $\lim_{s \rightarrow 0} s \frac{1}{s^2} \frac{1}{1+KG(s)} = \frac{1}{(2.34)(0.25)} = 1.7$ meters. (1)

- b) No, it is not possible to increase the type number – at least not by adding another integrator with no further modifications. If there are two poles at the origin, the root locus will have asymptotes at $\pm \frac{1}{3}\pi$, which would cause these poles to immediately cross into the RHP, so the closed-loop system will always be unstable.

Question 3 [18 marks]

We will now consider proportional control of the same drone system as in question 2, but from a frequency response perspective. The inverse Nichols chart of the system is shown on the next page.

- a) Calculate the gain required for the system to track a ramp with the accuracy specified in question 2. Roughly sketch the new response at this gain on the chart. [1 mark]

If this gain is applied,

- Will the system remain stable if a sensor delay causes the phase at around $\omega =$ to be reduced by 30 degrees? Explain. [3 marks]
- Determine the damping ratio, and hence percentage overshoot, of the closed-loop system. [2 marks]
- Are there any frequencies of external disturbances that the closed-loop system will amplify? If so, give a rough value of the frequencies that will be most affected, and how much they will be amplified by. [3 marks]

- b) Consider the following specifications (with the original gain of the system):

- The percentage overshoot specification in question 2
- Sensitivity $|S|$ at $\omega = 1$ rad/s of ≤ 3 dB
- Sensitivity $|S|$ for all frequencies of ≤ 2 dB

For each specification, indicate the boundary on the inverse Nichols chart, and determine the maximum increase in the system's gain – and hence, the minimum tracking error (in meters) for a ramp input – that is possible without violating that specification. [3 x 3 marks]

Solutions:

- a) In the previous question, it was found that $K \geq 40$ (32 dB) was required. The response with the gain at all frequencies increased by 32 dB is shown on the chart.

- No, the phase margin is < 30 degrees so this would cause the system to become unstable.
- By inspection, the phase margin is somewhere between 10 and 20 degrees. Let's call it 15. The closed-loop DC gain will be 1, because the system is type 1, so we can use the rule of thumb $\zeta = 0.01\Phi_m = 0.01(15) = 0.15$.

This leads to a %OS of $100e^{-\frac{0.15\pi}{\sqrt{1-0.15^2}}} = 62.1\%$

- Yes. The response passes through sensitivity circles with a magnitude >1 (0 dB) so the system will magnify some disturbances. The largest circle intersected is 10 dB at 2 rad/s, so disturbances around this frequency will be magnified by a gain of 3.16.

- b) i) We previously found that the damping ratio must be >0.46 to satisfy the %OS specification. With a closed-loop gain of 1, this gives a phase margin of 46 degrees. This is plotted on the inverse Nichols chart.

The frequency response is about 46 degrees away from -1 at a point with a magnitude of around -9 dB, so the gain can be increased by 9 dB (2.81) without violating this specification. This gives

tracking accuracy to $\lim_{s \rightarrow 0} s \frac{1}{s^2} \frac{1}{1+KG(s)} = \frac{1}{(2.81)(0.25)} = 1.4$ meters.

- ii) This specification requires that the point corresponding to 1 rad/s be kept out of the 3 dB sensitivity circle. This point is about 9 dB below this circle, so the gain can be increased by 9 dB.

This is the same value as in b.i, and gives tracking accuracy to 1.4 meters.

iii) This specification requires that the whole response does not cross into the 2 dB sensitivity circle. The response is currently on the boundary of this circle, so the gain cannot be increased at all (so, $K = 1$). This gives tracking accuracy to within $\frac{1}{0.25} = 4$ meters.

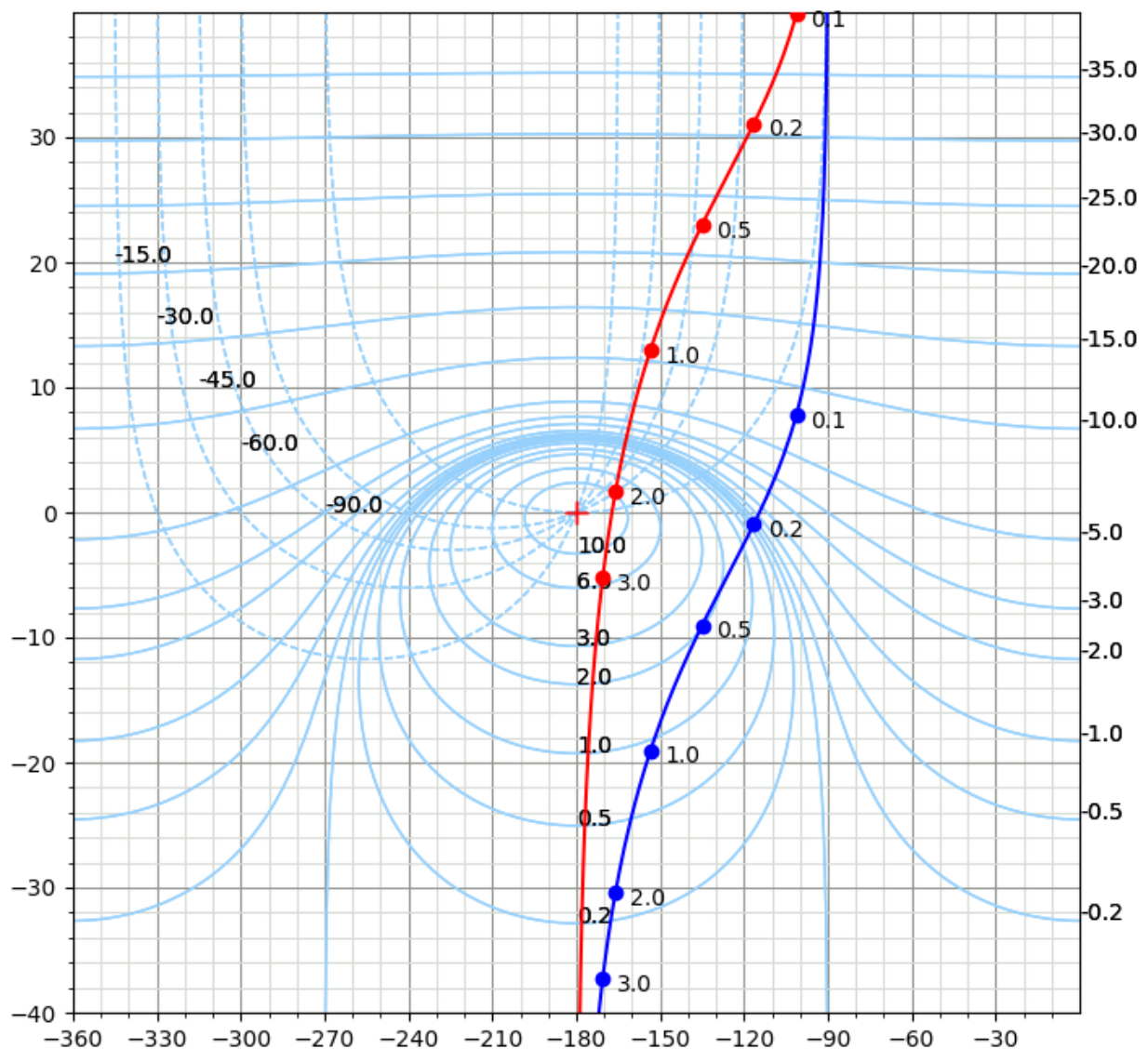


Figure 4: Inverse Nichols chart with gain increased as in part a

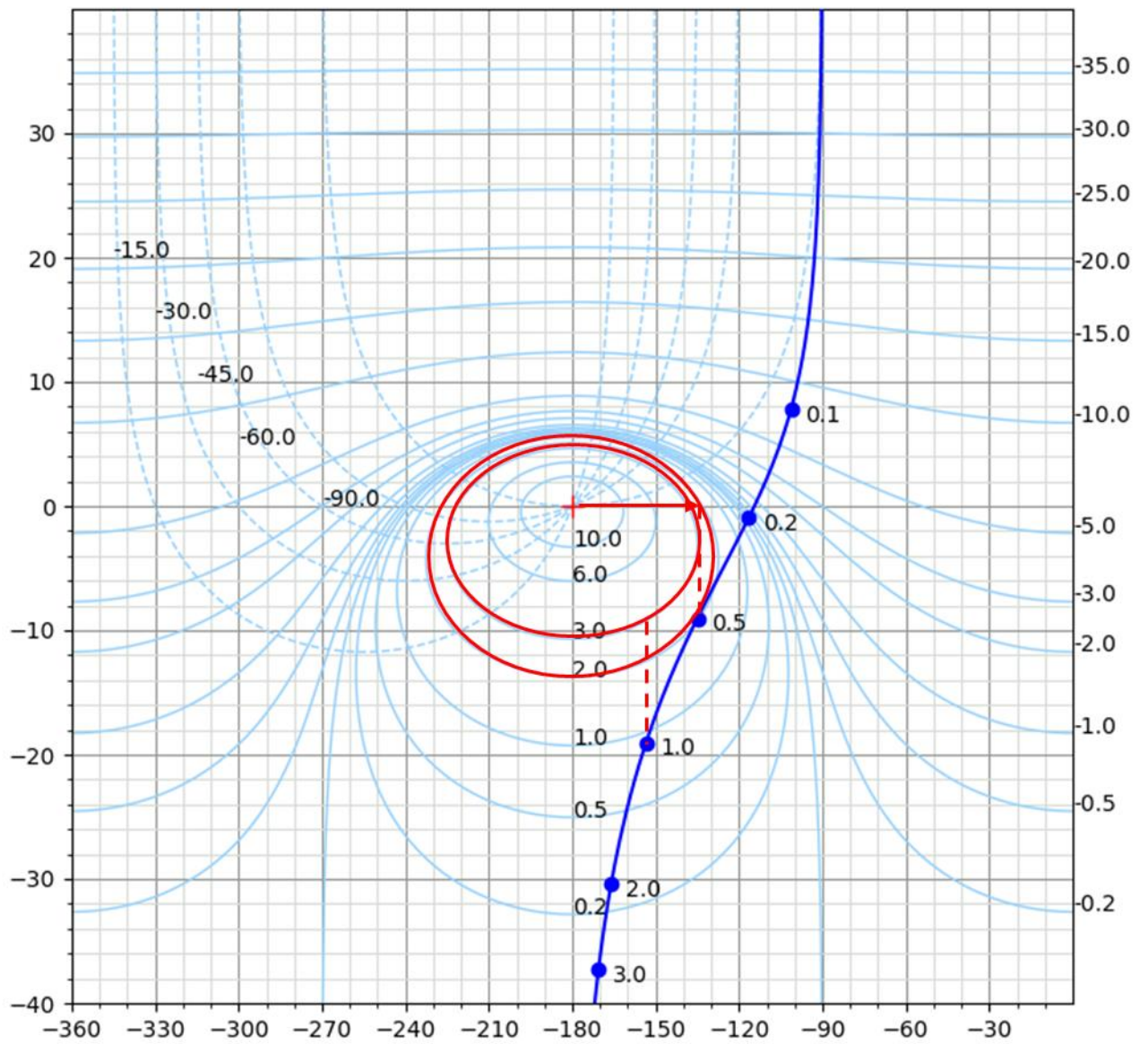


Figure 5: Inverse nichols chart with specification boundaries from part b